

1. Labels, relations, and formulas

- Labels: $\langle w, v, u, \dots \rangle$ range over worlds.
- Relational atoms:
 - $\langle w R^{\{\text{Rec}\}}_e v \rangle$: $\langle v \rangle$ is $\langle e \rangle$ -recognition-accessible from $\langle w \rangle$
 - $\langle w R^{\{\text{Comp}\}}_e v \rangle$: $\langle v \rangle$ is $\langle e \rangle$ -comprehension-accessible from $\langle w \rangle$
 - $\langle \text{Prev}(v, w) \rangle$: $\langle v \rangle$ is the immediate predecessor of $\langle w \rangle$ (for $\langle P \rangle$)
 - $\langle \text{Next}(w, v) \rangle$ or $\langle \text{Fut}(w, v) \rangle$: $\langle v \rangle$ is a future world of $\langle w \rangle$ (for $\langle F \rangle$)
- Labelled formulas: $\langle w : \varphi \rangle$, where $\langle \varphi \rangle$ is any formula of the language:
 - Atoms: $\langle \text{Model} \rangle$, $\langle \text{Predict} \rangle$, $\langle \text{Track} \rangle$, $\langle \text{SurpriseHigh} \rangle$, $\langle \text{SurpriseLow} \rangle$, $\langle \text{Notice} \rangle$, $\dots$
 - Connectives: $\langle \neg, \wedge, \vee, \rightarrow \rangle$
 - Modal: $\langle \Box^{\{\text{Rec}\}} \varphi \rangle$, $\langle \Box^{\{\text{Comp}\}} \varphi \rangle$
 - Temporal: $\langle P \varphi \rangle$, $\langle F \varphi \rangle$

2. Sequents

A sequent is of the form:

$$\frac{\Gamma}{\Delta}$$

where:

- $\langle \Gamma \rangle$ is a finite set of relational atoms (e.g. $\langle w R^{\{\text{Rec}\}}_e v \rangle$, $\langle \text{Prev}(v, w) \rangle$, ...)
- $\langle \Delta \rangle$ are finite multisets of labelled formulas $\langle w : \varphi \rangle$.

Intuition: under relational assumptions $\langle \Gamma \rangle$, from all formulas in $\langle \Delta \rangle$ we can derive at least one formula in $\langle \Delta \rangle$.

3. Structural and propositional rules

Identity:

$$\frac{\Gamma}{\Gamma, w : \varphi} \rightarrow \frac{\Gamma, w : \varphi}{\Delta, w : \varphi}$$

Weakening, contraction, cut as usual, acting on labelled formulas and relational atoms.

Propositional rules are standard, acting pointwise on labels. For example:

- $\neg(\wedge L)$:

$$\frac{\Gamma, w:\varphi \rightarrow \Delta}{\Gamma, w:(\varphi \wedge \psi) \rightarrow \Delta}$$

- $\neg(\rightarrow R)$:

$$\frac{\Gamma, w:\varphi \rightarrow \Delta, w:\psi}{\Gamma \rightarrow \Delta, w:(\varphi \rightarrow \psi)}$$

(and similarly for $\neg$ and $\vee$).

4. Modal rules (labelled K for each box)

4.1 Recognition modality $\Box^{\text{Rec}}_e$

- $\neg(\Box^{\text{Rec}}_e R)$ (Right-introduction):

$$\frac{\Gamma, w \in R^{\text{Rec}}_e v, \Gamma \rightarrow \Delta, v:\varphi}{\Gamma \rightarrow \Delta, w:\Box^{\text{Rec}}_e \varphi}$$

with the side condition: v is fresh in the conclusion (not occurring there).

- $\neg(\Box^{\text{Rec}}_e L)$ (Left-introduction):

$$\frac{\Gamma, w \in R^{\text{Rec}}_e v, \Gamma, v:\varphi \rightarrow \Delta}{\Gamma, w:\Box^{\text{Rec}}_e \varphi \rightarrow \Delta}$$

4.2 Comprehension modality $\Box^{\text{Comp}}_e$

Exactly parallel:

- $\neg(\Box^{\text{Comp}}_e R)$:

$$\frac{\Gamma, w \in R^{\text{Comp}}_e v, \Gamma \rightarrow \Delta, v:\varphi}{\Gamma \rightarrow \Delta, w:\Box^{\text{Comp}}_e \varphi}$$

- $\Box^{\text{Comp}}_e L$:

$$\frac{\{ \text{mathcal{R}}, w R^{\text{Comp}}_e v, \Gamma, v:\varphi \rightarrow \Delta \}}{\{ \text{mathcal{R}}, \Gamma, w:\Box^{\text{Comp}}_e \varphi \rightarrow \Delta \}}$$

These are standard labelled K-rules, one pair per modality.

5. Temporal rules (for P and F)

We use $\text{Prev}(v,w)$ and $\text{Fut}(w,v)$ in $\text{mathcal{R}}$.

5.1 Past operator P

- $(P R)$:

$$\frac{\{ \text{mathcal{R}}, \text{Prev}(v,w), \Gamma \rightarrow \Delta, v:\varphi \}}{\{ \text{mathcal{R}}, \Gamma \rightarrow \Delta, w:P\varphi \}}$$

- $(P L)$:

$$\frac{\{ \text{mathcal{R}}, \text{Prev}(v,w), \Gamma, v:\varphi \rightarrow \Delta \}}{\{ \text{mathcal{R}}, \Gamma, w:P\varphi \rightarrow \Delta \}}$$

5.2 Future operator F

- $(F R)$:

$$\frac{\{ \text{mathcal{R}}, \text{Fut}(w,v), \Gamma \rightarrow \Delta, v:\varphi \}}{\{ \text{mathcal{R}}, \Gamma \rightarrow \Delta, w:F\varphi \}}$$

- $(F L)$:

$$\frac{\{ \text{mathcal{R}}, \text{Fut}(w,v), \Gamma, v:\varphi \rightarrow \Delta \}}{\{ \text{mathcal{R}}, \Gamma, w:F\varphi \rightarrow \Delta \}}$$

(You can constrain $\backslash(\text{mathsf{Prev}})\backslash$ and $\backslash(\text{mathsf{Fut}})\backslash$ with extra relational rules if you want specific temporal frames.)

6. Domain axioms as initial sequents

For each entity $\backslash(e)$, we encode the recognition/comprehension invariants as axiom schemata (initial sequents) at every label $\backslash(w)$:

- Comprehension constraints:

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{mathsf{Model}}e\text{big})\backslash$
- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{mathsf{Predict}}e\text{big})\backslash$
- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{mathsf{Track}}e\text{big})\backslash$
- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{mathsf{SurpriseLow}}e\text{big})\backslash$

- Recognition constraints:

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Rec}}})e \text{ top } \text{rightarrow} (\text{neg } \text{mathsf{Model}}e \text{ lor } \text{neg } \text{mathsf{Predict}}_e)\text{big})\backslash$
- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Rec}}})e \text{ top } \text{rightarrow} (\text{mathsf{SurpriseHigh}}e \text{ lor } \text{neg } \text{mathsf{Track}}_e)\text{big})\backslash$

- Mutual incompatibility (optional):

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{neg } \text{Box}^{\text{mathsf{Rec}}})e \text{ top } \text{big})\backslash$

- Noticing invariant:

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{mathsf{Notice}}e \text{rightarrow} \text{mathsf{SurpriseHigh}}e\text{big})\backslash$
- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{mathsf{Notice}}e \text{rightarrow } P(\text{neg } \text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{big})\backslash$

- No noticing under low surprise (optional):

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{mathsf{SurpriseLow}}e \text{rightarrow} \text{neg } \text{mathsf{Notice}}e\text{big})\backslash$

From these, you can build explicit proof trees showing, for example:

- $\backslash(\text{mathcal{R}}, \backslash \backslash, \text{Rightarrow}; w:\text{big}(\text{Box}^{\text{mathsf{Comp}}})e \text{ top } \text{rightarrow} \text{neg } \text{mathsf{Notice}}e\text{big})\backslash$

- and, via the (P) -rules, that $(w:\mathsf{Notice}e)$ entails some (v) with $(\mathsf{Prev}(v,w))$ and $(v:\neg \Box^{\mathsf{Comp}}e \text{ top})$.

If you want, next step we can walk through one concrete labelled proof tree for a statement like:

$$\begin{array}{c} \frac{}{\vdash} \\ \vdash \text{Rightarrow} w:\mathsf{big}(\mathsf{Notice}e \rightarrow P(\neg \Box^{\mathsf{Comp}}e \text{ top}))\mathsf{big} \\ \vdash \end{array}$$

so you can see the system “move” through labels and relations..

Propositional rules are standard, acting pointwise on labels. For example:

- $(\land L)$:

$$\begin{array}{c} \frac{}{\vdash} \\ \frac{\frac{}{\vdash} \Gamma, w:\varphi \vdash \psi \rightarrow \Delta}{\frac{}{\vdash} \Gamma, w:(\varphi \land \psi) \vdash \Delta}}{\vdash} \end{array}$$

- $(\rightarrow R)$:

$$\begin{array}{c} \frac{}{\vdash} \\ \frac{\frac{}{\vdash} \Gamma, w:\varphi \vdash \Delta, w:\psi}{\frac{}{\vdash} \Gamma \vdash \Delta, w:(\varphi \rightarrow \psi)}}{\vdash} \end{array}$$

(and similarly for $(\neg, \lor)$).

4. Modal rules (labelled K for each box)

4.1 Recognition modality $(\Box^{\mathsf{Rec}}_e)$

- $(\Box^{\mathsf{Rec}}_e R)$ (Right-introduction):

$$\begin{array}{c} \frac{}{\vdash} \\ \frac{\frac{}{\vdash} w R^{\mathsf{Rec}}_e v, \Gamma \vdash \Delta, v:\varphi}{\frac{}{\vdash} \Gamma \vdash \Delta, w:\Box^{\mathsf{Rec}}_e \varphi}}{\vdash} \end{array}$$

with the side condition: (v) is fresh in the conclusion (not occurring there).

- $(\Box^{\mathsf{Rec}}_e L)$ (Left-introduction):

$$\frac{}{\vdash}$$

$$\frac{\{\mathcal{R}, \Box w R^{\Box} v, \Gamma, \varphi \rightarrow \Delta\}}{\{\mathcal{R}, \Gamma, \Box w \Box^{\Box} \varphi \rightarrow \Box \varphi \rightarrow \Delta\}}$$

4.2 Comprehension modality $\Box^{\Box}$

Exactly parallel:

- $\Box^{\Box} R$:

$$\frac{\{\mathcal{R}, \Box w R^{\Box} v, \Gamma \rightarrow \Delta, \varphi\}}{\{\mathcal{R}, \Gamma \rightarrow \Delta, \Box w \Box^{\Box} \varphi \rightarrow \Box \varphi\}}$$

- $\Box^{\Box} L$:

$$\frac{\{\mathcal{R}, \Box w R^{\Box} v, \Gamma, \varphi \rightarrow \Delta\}}{\{\mathcal{R}, \Gamma, \Box w \Box^{\Box} \varphi \rightarrow \Box \varphi \rightarrow \Delta\}}$$

These are standard labelled K-rules, one pair per modality.

5. Temporal rules (for P and F)

We use $\Box^{\Box} \text{Prev}(v, w)$ and $\Box^{\Box} \text{Fut}(w, v)$ in $\mathcal{R}$.

5.1 Past operator P

- $P R$:

$$\frac{\{\mathcal{R}, \Box^{\Box} \text{Prev}(v, w), \Gamma \rightarrow \Delta, \varphi\}}{\{\mathcal{R}, \Gamma \rightarrow \Delta, \Box^{\Box} P \varphi \rightarrow \Box \varphi\}}$$

- $P L$:

$$\frac{\{\mathcal{R}, \Box^{\Box} \text{Prev}(v, w), \Gamma, \varphi \rightarrow \Delta\}}{\{\mathcal{R}, \Gamma, \Box^{\Box} P \varphi \rightarrow \Box \varphi \rightarrow \Delta\}}$$

5.2 Future operator F

- $\forall (F R) \forall$:

$$\forall \left[\frac{\text{mathcal{R}}, \text{mathsf{Fut}}(w, v), \Gamma \rightarrow \Delta, v: \varphi}{\text{mathcal{R}}, \Gamma \rightarrow \Delta, w: F \varphi} \right]$$

- $\forall (F L) \forall$:

$$\forall \left[\frac{\text{mathcal{R}}, \text{mathsf{Fut}}(w, v), \Gamma, v: \varphi \rightarrow \Delta}{\text{mathcal{R}}, \Gamma, w: F \varphi \rightarrow \Delta} \right]$$

(You can constrain $\text{mathsf{Prev}}$ and $\text{mathsf{Fut}}$ with extra relational rules if you want specific temporal frames.)

6. Domain axioms as initial sequents

For each entity e , we encode the recognition/comprehension invariants as axiom schemata (initial sequents) at every label w :

- Comprehension constraints:

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Comp}}} e \top \rightarrow \text{mathsf{Model}} e \big)$

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Comp}}} e \top \rightarrow \text{mathsf{Predict}} e \big)$

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Comp}}} e \top \rightarrow \text{mathsf{Track}} e \big)$

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Comp}}} e \top \rightarrow \text{mathsf{SurpriseLow}} e \big)$

- Recognition constraints:

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Rec}}} e \top \rightarrow (\neg \text{mathsf{Model}} e \vee \neg \text{mathsf{Predict}}_e) \big)$

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Rec}}} e \top \rightarrow (\text{mathsf{SurpriseHigh}} e \vee \neg \text{mathsf{Track}}_e) \big)$

- Mutual incompatibility (optional):

- $\text{mathcal{R}}, \Gamma \rightarrow w: \Box^{\text{mathsf{Comp}}} e \top \rightarrow \neg \Box^{\text{mathsf{Rec}}} e \big)$

- Noticing invariant:

- $\vdash \text{Notice} \rightarrow \text{SurpriseHigh}$
- $\vdash \text{Notice} \rightarrow \neg \text{Comp}$

- No noticing under low surprise (optional):

- $\vdash \text{SurpriseLow} \rightarrow \text{Notice}$

From these, you can build explicit proof trees showing, for example:

- $\vdash \neg \text{Comp} \rightarrow \neg \text{Notice}$
- and, via the $\vdash$ -rules, that $\vdash \text{Notice}$ entails some $\vdash$ with $\vdash \text{Prev}(v,w)$ and $\vdash \neg \text{Comp}$.

If you want, next step we can walk through one concrete labelled proof tree for a statement like:

$$\vdash \text{Notice} \rightarrow \neg \text{Comp}$$

so you can see the system “move” through labels and relations.